

Investigation and Dependency Analysis of Silicon Film Thickness on Performance of Surrounding Gate MOSFET at Subthreshold Regime

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Abstract

The characteristics of cylindrical gate all around MOSFET have been studied in terms of the dependence of various device performance metrics on silicon film thickness (T_{si}). In this work, the effect of silicon film thickness on numerous performance metrics like threshold voltage (V_t), On current (I_{on}), subthreshold leakage current (I_{off}), On-Off current ratio (I_{on}/I_{off}) and DIBL of cylindrical GAA are comprehensively evaluated and analysed. It has been observed that cylindrical gate all around MOSFET offer high drive-current (I_{on}), low leakage current (I_{off}), superior On-Off current ratio and hence enhanced gain. Thus, for ultra-low power applications surrounding nanowire contemplated as superior aspect. In the present study, all the device performances are investigated through ATLAS device simulator from Silvaco.

Keywords: Cylindrical gate all around (CGAA), Drain Induced Barrier Lowering (DIBL), Short channel effects (SCE)

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INTRODUCTION

Scaling the base component measure directly affects short channel effects (SCE), for example effect of charge sharing close to source/deplete transporter expansion, drain-induced barrier lowering and punch through effects. For SOI advances, scaling additionally requires a diminishment of both the silicon film and buried oxide thickness with a specific end goal to decrease the SCE [1]. This will impact on the threshold voltage, which as indicated by the traditional hypothesis is relied upon to be brought down for reduced gate lengths and film thickness [2]. The developing interest of economical gadgets with complex functionalities and truculent scaling in package density and execution dependably move toward becoming necessities. In turns, the ceaseless growth in the framework necessities requests more elevated amount of integration and execution which have been comprehended by persistent downscaling of MOS devices. However, MOS transistor scaling has resulted in unacceptable leakage currents beyond

subthreshold region [3, 4]. In most recent couple of years, analyst's shows solid consideration and enthusiasm towards surrounded nanowire MOSFET rather than planner structure MOS [3, 5]. Cylindrical gate all around (GAA) MOSFET is a noteworthy innovation and most encouraging contender to supplant traditional MOSFET, fit into people to come [6–9].

Scaling silicon film thickness is desirable for better short channel behavior and reduced floating body effect [10–11]. However, this can make the characteristic of edge transistor becomes very different from moderate thin-film devices. In this paper, we have demonstrated that scaling the silicon film thickness has strong impact on the threshold voltage and subthreshold leakage because of edge transistors [12]. In this study the effect of silicon film thickness on various electrical characteristics like threshold voltage (V_t), On current (i_{on}), subthreshold leakage current (I_{off}), On-Off current ratio (I_{on}/I_{off}) and DIBL are explored.

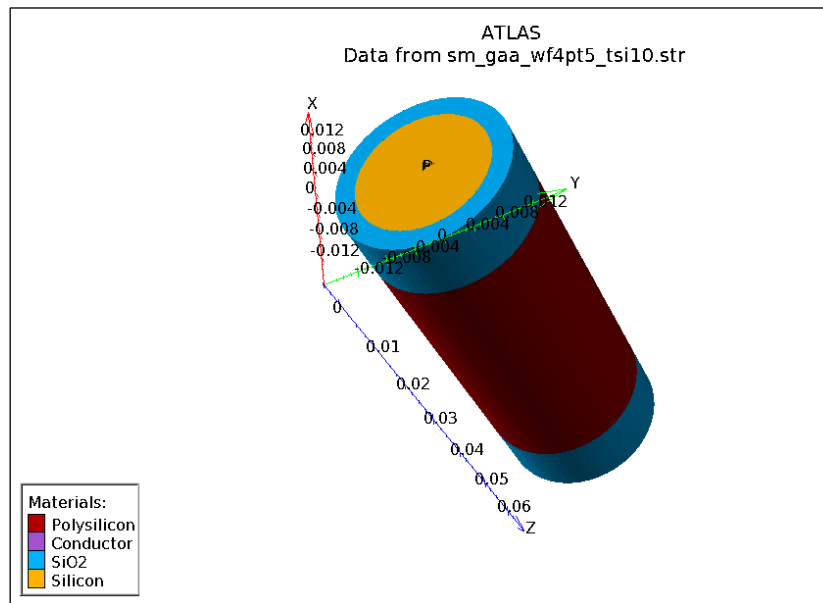


Fig. 1: Schematic Structure of Cylindrical Gate all Around (GAA) MOSFET.

Device Structure and Specification

The schematic diagram of the cylindrical gate all around (CGAA) MOSFET structures used for simulation is shown in Figure 1. This structure is implemented using Atlas Silvaco tool. The radial directions are assumed to be along radius and lateral direction along z-axis of the cylinder as shown in Figure 1.

The details of device physical parameters used in the structure are shown in Table 1. SOI substrate in cylindrical gate all around (GAA) MOSFETs having n-type channels. The Length of the gate (L_g) is 45 nm and radius of 5 nm. Gate oxide thickness of SiO_2 (t_{ox}) is 1 nm.

Table 1: Device Dimensions and Dopings.

Parameter	Value
Gate Length (L_g)	45nm
Oxide Thickness (t_{ox})	1nm
Channel Doping	$1.0 \times 10^{18} \text{ cm}^{-3}$
Source Doping (N_D)	$1.0 \times 10^{20} \text{ cm}^{-3}$
Drain Doping (N_D)	$1.0 \times 10^{20} \text{ cm}^{-3}$

The source/drain extension uniform doping profile was set as Gaussian with the peak concentration of $1.0 \times 10^{20} \text{ cm}^{-3}$. We investigated threshold voltage (V_t), On Current (I_{ON}), Off Current (I_{OFF}), On-Off current (I_{ON}/I_{OFF}) ratio and DIBL variation for the film

thickness ranging from 2.5 to 10 nm in the present simulation study.

SIMULATION RESULTS

Transfer Characteristics

Figure 2 shows the drain current I_{ds} versus gate voltage V_{gs} characteristics of cylindrical gate all around MOSFET for different values of silicon film thickness. The silicon film thickness has been varied from 2.5 nm to 10 nm.

The results show that the subthreshold behaviour of device advances, as the film thickness is decreased to lower values. It is because as the film thickness decreases, the corresponding threshold voltage increases, which in turn decreases the off-state leakage current and results in the enhancement in the device performance. This is very desirable characteristics of the device for various applications.

Threshold Voltage Variation with Film Thickness

We investigated the threshold voltage (V_t) variation as a function of film thickness ranging from 2 nm to 10 nm in the present simulation study.

For CGAA MOSFET threshold voltage can be expressed as

$$V_{th} = V_{fb} + \phi_{smin-th} + \frac{\lambda q N_a}{4\epsilon_{si}} \left(1 - \frac{t_{si}^2}{\lambda}\right)$$

Where,

V_{fb} is flat band voltage, $\phi_{smin-th}$ is the value of surface potential at which the volumetric inversion electron charge density in the Si device is equal to the body doping.

$$\lambda = t_{si}^2 \left(1 + \frac{2\epsilon_{si}t_{ox}}{\epsilon_{ox}t_{si}}\right)$$

ϵ_{ox} and ϵ_{si} are the permittivity of SiO_2 and Si respectively.

It has been analyzed that by decreasing the film thickness of cylindrical gate all around MOSFET, the corresponding threshold voltage increases as shown in Figure 3. The results as shown in this Figure 2 have been obtained for a device having $L_g = 45 \text{ nm}$, $R = t_{si}/2 = 5 \text{ nm}$, $t_{ox} = 1 \text{ nm}$. This increase in threshold voltage as we decrease in film thickness is because of rise of the minimum energy for holes within the valence band. It is well known that inversion carrier concentration required to achieve threshold voltage is increased as film thickness is decreased.

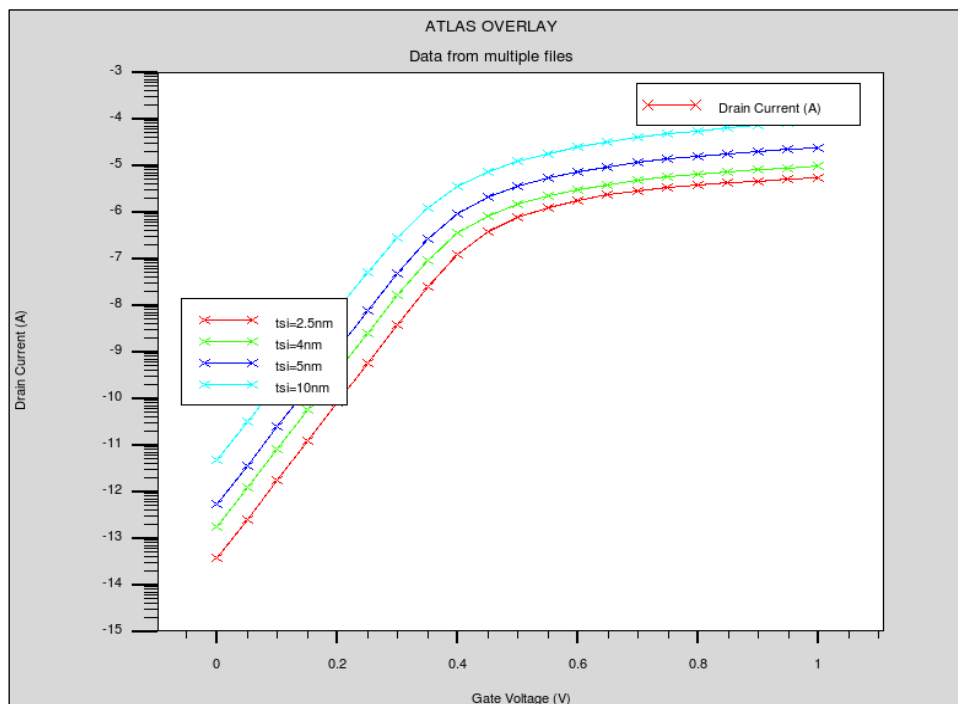


Fig. 2: I_{ds} versus V_{gs} (Transfer characteristics) Curve for Different Film Thickness.

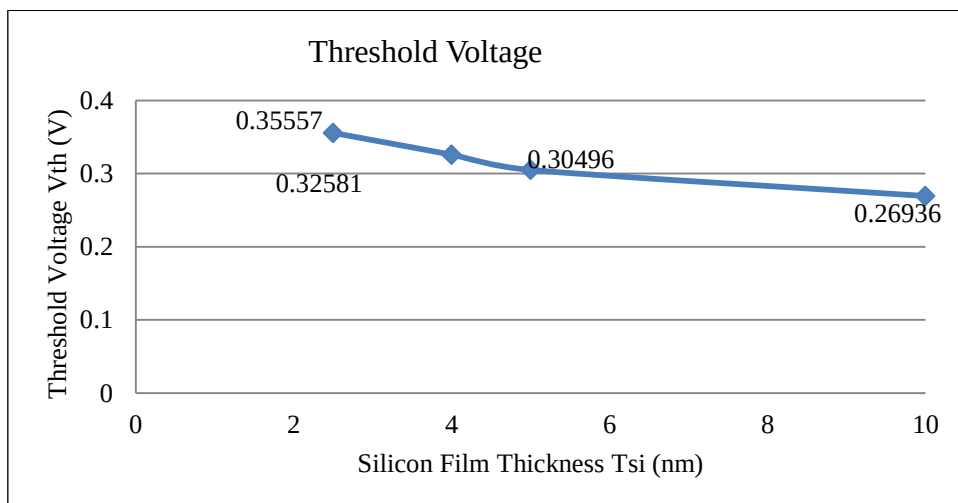


Fig. 3: Threshold Voltage as Function of Silicon Film Thickness.

On-Current Variation with Film Thickness

The device on-current behavior as a function of silicon film thickness has been illustrated in Figure 4. It is clear that device on-current is suffered as threshold voltage increased with decrease in film thickness of cylindrical gate all around MOSFET structure.

Off-Current Variation with Film Thickness

The device off-current behavior as a function of film thickness has been illustrated in Figure 5.

It is clear that device off-current also decreases as decrease in film thickness of cylindrical gate all around MOSFET structure.

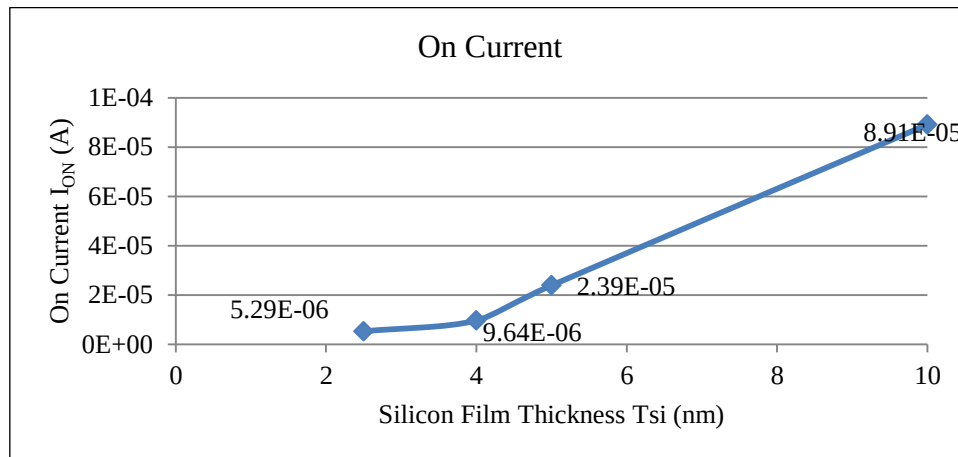


Fig. 4: On-Current Behaviour as a Function of Silicon Film Thickness.

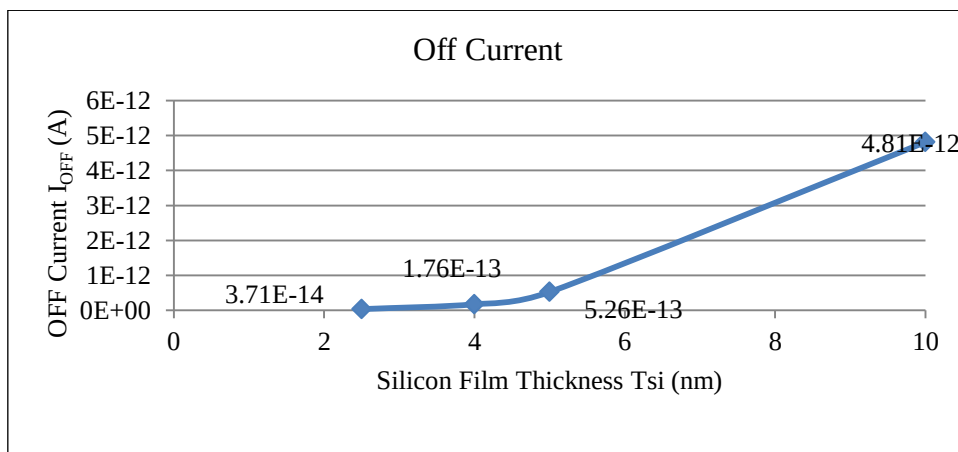


Fig. 5: Leakage Current Analysis as a Function of Silicon Film Thickness.

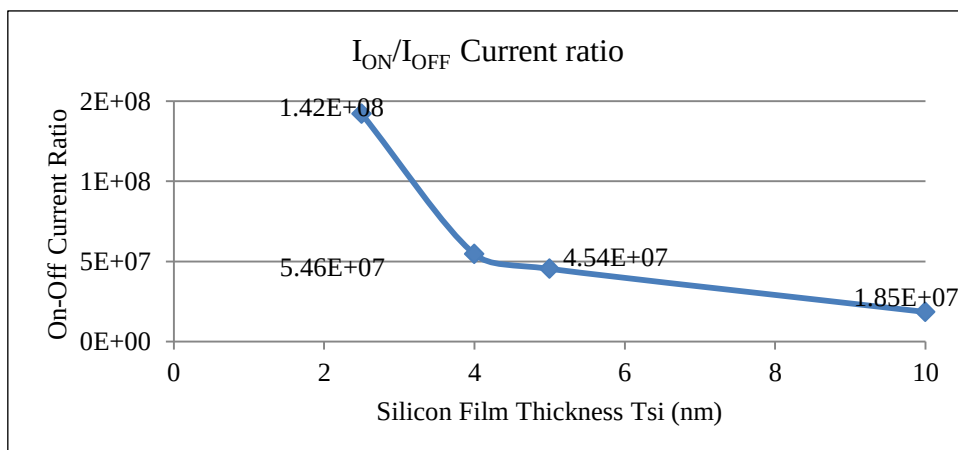


Fig. 6: I_{ON}/I_{OFF} Current Ratio Variation as a Function of Silicon Film Thickness.

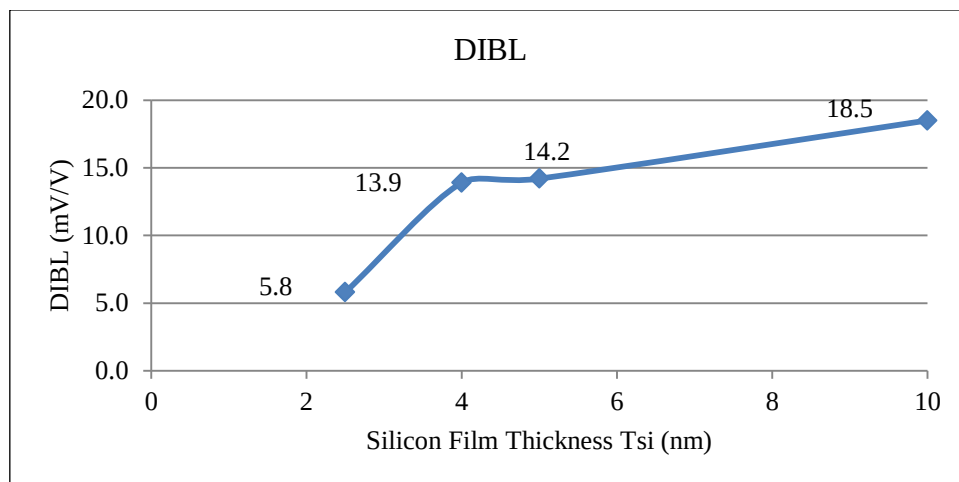


Fig. 7: DIBL Variation as a Function of Silicon Film Thickness.

I_{ON}/I_{OFF} Current Ratio Variation with Film Thickness

The device I_{ON}/I_{OFF} current ratio behavior as a function of silicon film thickness has been illustrated in Figure 6. It is clear that device I_{ON}/I_{OFF} current ratio increases as decrease in silicon film thickness of cylindrical gate all around MOSFET structure.

DIBL Variation with Film Thickness

The DIBL behavior as a function of silicon film thickness has been illustrated in Figure 7.

The drain induced barrier lowering (DIBL) effect is defined as the lateral shift of the transfer curves in the subthreshold regime when the drain voltage varies from 0.05 V to 1 V. The Result of Simulation were summarised in Table 2.

Table 2: Simulated value of Cylindrical Gate all Around MOSFET's Electrical Characteristics for Different Values of Silicon Film Thickness of the Structure.

Silicon Film Thickness Tsi (nm)	I_{ON} (A)	I_{OFF} (A)	I_{ON}/I_{OFF}	Vth (V)	DIBL (mV/V)
2.5	5.29 E-6	3.71 E-14	1.42 E8	0.36	5.8
4	9.64 E-6	1.76 E-13	5.46 E7	0.33	13.9
5	2.39 E-5	5.26 E-13	4.54 E7	0.31	14.2
10	8.91 E-5	4.81 E-12	1.85 E7	0.27	18.5

CONCLUSION

It has been analysed that device has higher driving current and leakage current for silicon film thickness 10 nm as matched to film thickness 5 nm, 4 nm and 2.5 nm. But the ratio of I_{ON}/I_{OFF} current increases by decreasing the silicon film thickness ranging from 10 nm to 2.5 nm. It is also observed Cylindrical GAA MOSFET having film thickness 2.5 nm has higher threshold voltage and lower DIBL effect. So, in this way we can advance the device performance by reducing the leakage current with variation of film thickness in sub-threshold regime.

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